

EXPERIMENT 21:

21. Develop a C program to implement worst fit algorithm of memory management.

PSEUDO CODE :

Start

Input memory blocks

Input process sizes

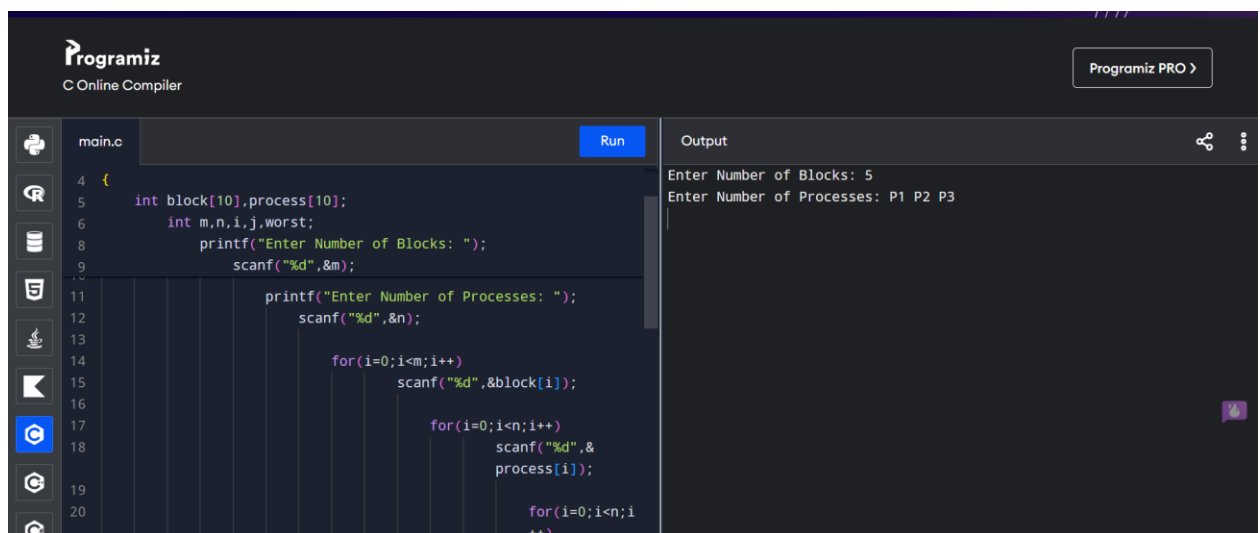
For each process

Find largest free block

Allocate process

Display allocation

Stop



The screenshot shows the Programiz Online C Compiler interface. On the left, a file explorer shows 'main.c'. The main editor displays the following C code:

```
1  //
2
3  #include <stdio.h>
4  {
5      int block[10], process[10];
6      int m, n, i, j, worst;
7      printf("Enter Number of Blocks: ");
8      scanf("%d", &m);
9
10     printf("Enter Number of Processes: ");
11     scanf("%d", &n);
12
13     for(i=0; i<m; i++)
14         scanf("%d", &block[i]);
15
16     for(i=0; i<n; i++)
17         scanf("%d", &process[i]);
18
19     for(i=0; i<n; i++)
```

The output window on the right shows the following text:

```
Enter Number of Blocks: 5
Enter Number of Processes: P1 P2 P3
```

CODE:

```
#include<stdio.h>
```

```
int main()

{

    int
    block[10],process[10];

    int m,n,i,j,worst;


    printf("Enter Number
of Blocks: ");

    scanf("%d",&m);


    printf("Enter Number
of Processes: ");

    scanf("%d",&n);


    for(i=0;i<m;i++)

scanf("%d",&block[i]);


    for(i=0;i<n;i++)

scanf("%d",&process[i])
;


    for(i=0;i<n;i++)

    {

        worst=-1;

        for(j=0;j<m;j++)
```

```

    {
    if(block[j]>=process[i])

        {

            if(worst==-1 ||
            block[j]>block[worst])

                worst=j;

            }

        }

    if(worst!=-1)

        {

printf("Process%d ->
Block

%d\n",i+1,worst+1);

        block[worst]-
        =process[i];

        }

    }

    return 0;

}

```